

Fibers crop insects

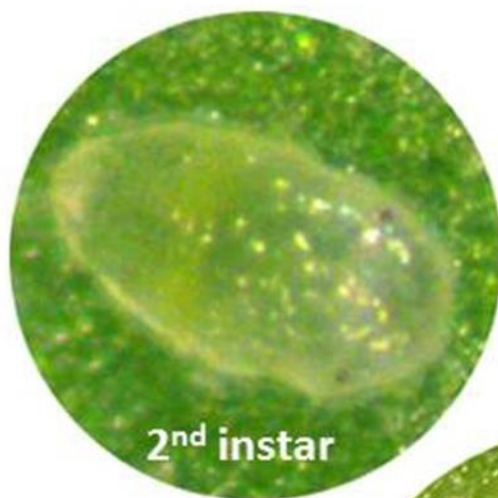
1- Silverleaf whitefly *Bemisia tabaci* (Gennadius) الذبابة البيضاء

Family : Aleyrodidae

Order : Hemiptera



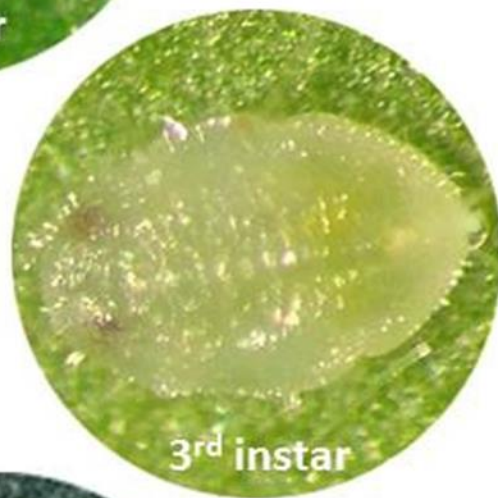
1st instar



2nd instar



Eggs



3rd instar



Adults



4th instar

Host range:

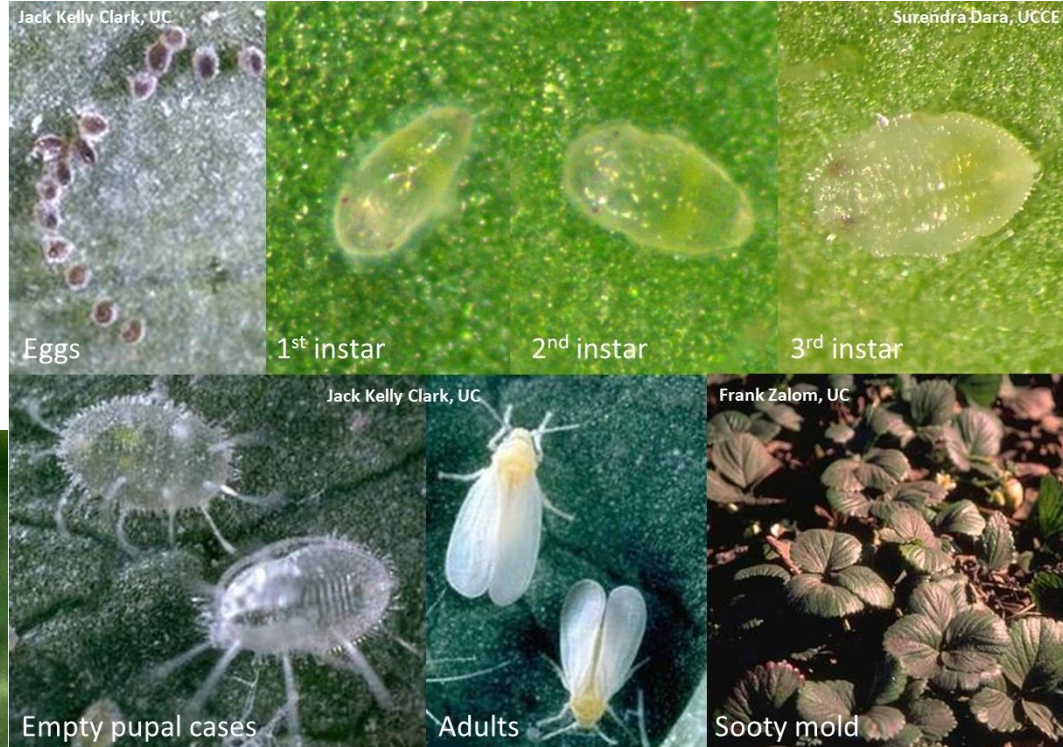
has been found on the following crop plants: avocado, broccoli, cauliflower, Chinese cabbage, cucumber, eggplant, fig, green bean, lettuce, soy bean, squash, sweetpotato, tomato, watermelon.

Harmful stage:

Adult and nymph.

Description:

The silver leaf whitefly adult is small, is about 0.9 to 1.2 mm in length, and holds its solid white wings roof-like over a pale-yellow body while at rest. The nymph look nothing like the adult. The nymphs are flattened, oval scalelike structures. Eggs are oblong and yellow, about 0.2 mm.



Damages:

The whitefly is recognized as an important pest on many crops. Three types of damage may be caused by the whitefly: 1) direct damage, 2) indirect damage and 3) virus transmission.

Direct feeding damage is caused by the piercing and sucking sap from the foliage of plants. This feeding causes weakening and early wilting of the plant and **reduces the plant growth which in turn influences the development of fruit and reduction in yield**. It may also cause leaf chlorosis, leaf withering, premature dropping of leaves and plant death. Infestations of whitefly nymphs are associated with the occurrence **of irregular ripening of tomatoes and silverleaf of squash.**



Indirect damage results by the accumulation of honeydew produced by the whiteflies. This honeydew serves as a substrate for the growth of black sooty mold on leaves and fruit. The mold reduces photosynthesis and lessens the market value of the plant or yields it unmarketable.

The third type of damage is caused by the vectoring of plant viruses by this insect. A small population of whiteflies is sufficient to cause considerable damage. Plant viruses transmitted by whiteflies cause over 40 diseases of vegetable and fiber crops worldwide.

In the past decade, whitefly-transmitted plant viruses have increased in prevalence and distribution. The recent impact has been devastating with yield losses ranging from 20 to 100 percent, depending upon the crop, season, and prevalence of the whitefly. Some diseases associated with whitefly include: **Lettuce necrotic yellows, irregular ripening of tomato, silver leaf of squash, cotton leaf curl, tobacco leaf curl.**

Chemical control



Closer 240 SC
Sulfoxaflor 240g/ L



Uphold 360 SC
Methoxyfenozide 300g/l +
Spinetoram 60 g/l

4- Cottonseed bug *Oxycarenus hyalinipennis* (Costa) بقة بذور القطن

Family : [Lygaeidae](#)

Order : [Hemiptera](#)

Host range:

Cotton seed bug, , is a serious pest of cotton and other plants in the cotton family. This pest can also feed on other fruits and seeds of unrelated plants causing significant damage.

Harmful stage:

nymph and Adult.



Description:

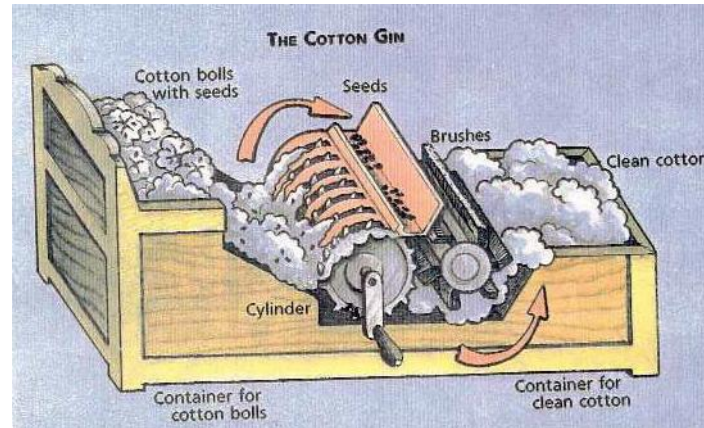
As the name suggests, this is a bug with transparent wings. The rest of the body is dark, giving the bug a contrasting black and white appearance. The head is shaped like the head of a rat , The bugs are 4-5 mm long. Nymphs can be reddish



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Damages:

These bugs are seed feeders. For practical purposes, if large numbers of small bugs are found feeding on cotton seeds (inside open bolls) or seeds of other malvaceous plants. Sometimes, the bugs are crushed in the ginning process, staining the lint. Greater damage is done to the seeds by reducing quality, germination and oil content (. Adult cotton seed bugs may feed on fruits and seeds of non-malvaceous plants, causing significant damage.

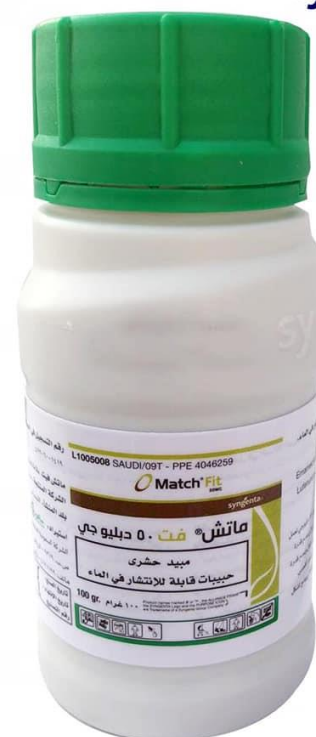


INSECTICIDAL



Radiant^{sc}
Spinetoram 120g\ L

syngenta



Match EC
Lufenuron 50g/ L

7- Cabbage Flea Beetles (*Phyllotreta cruciferae*) الخنفساء البرغوثية

Family : Chrysomelidae

Order : Coleoptera

Host range:

The crucifer flea beetle has a narrow host range restricted to plants primarily in the mustard family (Cruciferae). all of the flea beetles prefer plant families that produce mustard oil, which is a known aggregation pheromone of the crucifer flea beetle. The most-preferred hosts are in the genus *Brassica* (Cruciferae), which include the major agricultural host attacked by flea beetle, Other hosts that flea beetles will attack in the garden setting are cabbage, turnip, cauliflower, kale, and radish.

Harmful stage:

Adult and Larvae.



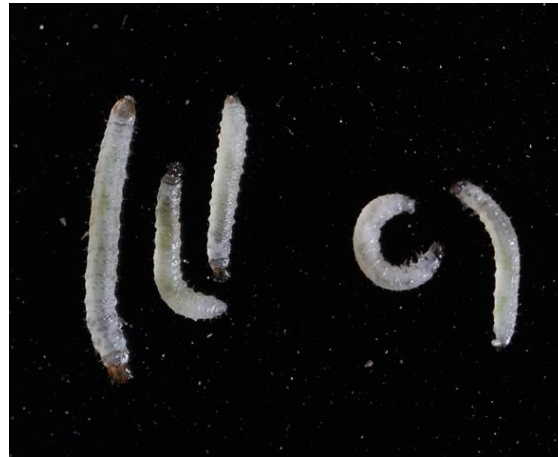
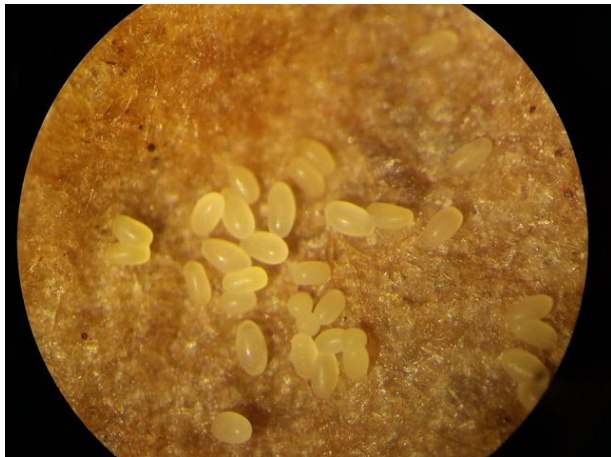
Description:

Crucifer flea beetle: The adult is a small, oval-shaped, blackish beetle with a bright blue sheen on the elytra, measuring about (2-3 mm) in length. Flea beetles have enlarged hind femora (thighs) on their hind legs, which they use to jump quickly when disturbed. Their name, flea beetle, arose from this behavior.

Eggs are yellow, oval, and about 0.38-0.46 mm long and deposited singly or in groups of three or four adjacent to the host plant's roots.

Larvae are small approximately 3 mm, whitish, slender, cylindrical worms. They have tiny legs and a brown head.

Pupae are similar in size to the adult and white in color except for the black eyes and the free body appendages, which are visible later in the pupal development.



Damages:

The adults feed on both the cotyledons and the young developing leaves giving the distinctive 'shot-holing', which affects the cotyledons and early leaves, resulting in stunting and poor plant vigor. If severe, the damage can kill the seedlings even before they emerge.



2- pink bollworm *Pectinophora gossypiella* دودة جوز القطن القرنفلية

Family : [Gelechiidae](#)

Order : [Lepidoptera](#)

Host range:

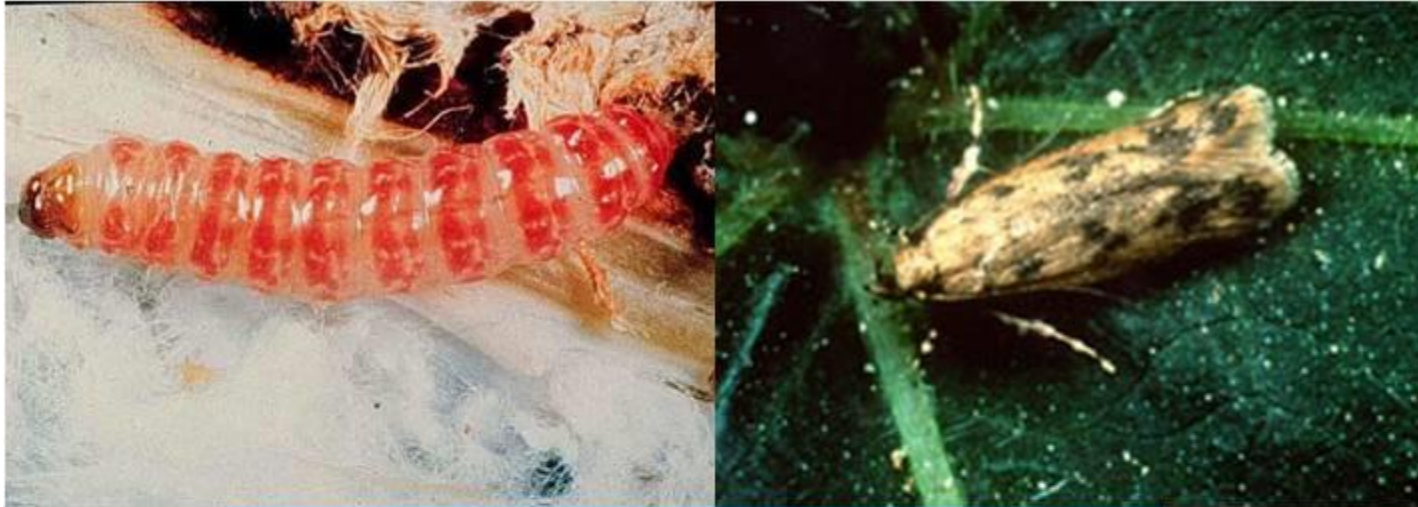
Cotton and okra

Harmful stage:

Larvae.

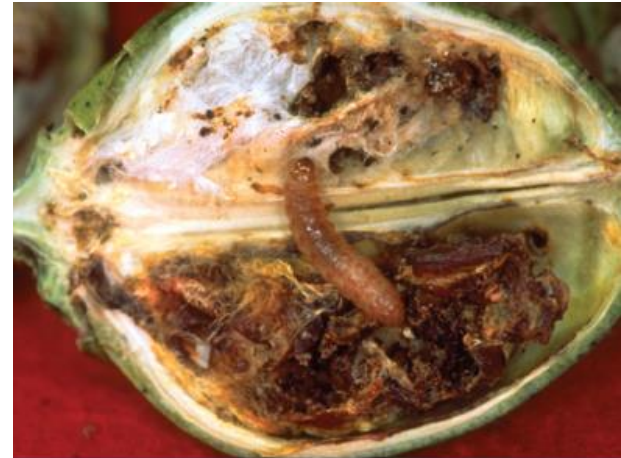
Description:

The body of the larva is up to 10 mm (0.39 inch) long. Young larvae are ivory-white, but they later develop pink bands; the last instar before pupation is reddish-pink. The adults have grayish-brown wings with a wingspan of about 20 mm (0.78 inch).



Damages:

The newly-hatched larvae feed externally on cotton bolls or flower buds for some hours, leaving typical marks. Then they bore themselves in, feed on the seed, and soil the lint with their frass. An individual larva may damage several bolls. The damaged bolls mature early and fail to open. After 2-3 weeks, the larva pupates inside the boll or emerges through a hole of approx. 2-3 mm diameter at its top, and then searches for a hiding-place either on the plant, or on the ground, where it will build its cocoon. Sometimes, several flower buds are spun together by numerous larvae, such that the petals cannot open properly.



Chemical Control



Coragen SC
Chlorantraniliprole 20%

Proclaim 05 sg
Emamectin benzoate 5%

Stimulating crop insects

1-**Common Mole Cricket** *Gryllotalpa gryllotalpa* (L.) الكاروب

Family : Gryllotalpidae

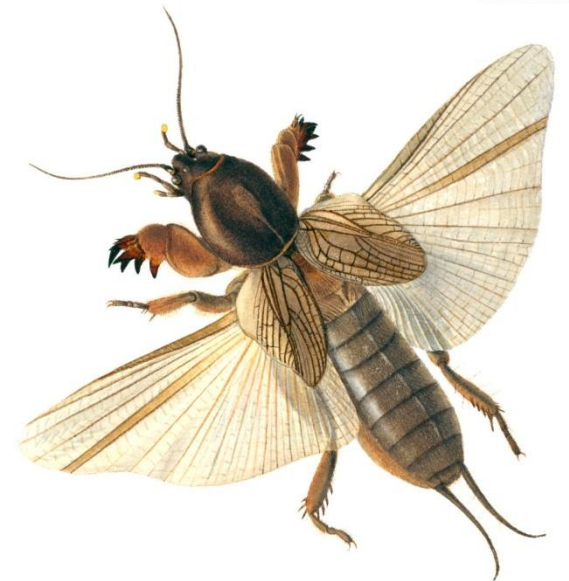
Order : Orthoptera

Host range:

Very general polyphage. Damages cereals, legumes, perennial grasses, potato, almost all vegetables, nearly all melons, nearly all gourds, beet, sunflower, tobacco, hemp, flax, and strawberry; it causes damage in hotbeds; in nurseries and young gardens, it damages vines, fruit trees, and many other tree species.

Harmful stage:

Adult and nymph



Description:

Length of body is 35-50 mm. Coloring brown with a silky shimmer, rather yellow on the underside, covered with thick, small hairs. Forelegs are of the digging type -- short, enlarged, with strong indentations. Hind legs have 3-4 inner spines. Prothorax elongate-ovate, 1.2-1.3 times longer than its maximum width. Elytra short, half the length of abdomen



Damages:

Larvae remain in their nest under the guard of female for 2-3 weeks. Feeding and burrowing holes underground, the Mole Cricket gnaws through the roots of plants, eating away tubers and rhizomes. It also feeds on many soil invertebrates, including insects and earthworms.. Mole crickets prefer light, fresh soils with high organic matter content where food is available and tunnels can be excavated easily. Occasionally, mole crickets wander above ground causing then extensive damage in plant beds.



Chemical control



Decis Expert EC
Deltamethrin 100g/ L



BESTOX 10_{EC}
(alpha-cypermethrin 10%)

